Lesson for the course "Human-Computer Interaction"

# Data collection and analysis: practical tips and hands-on activities

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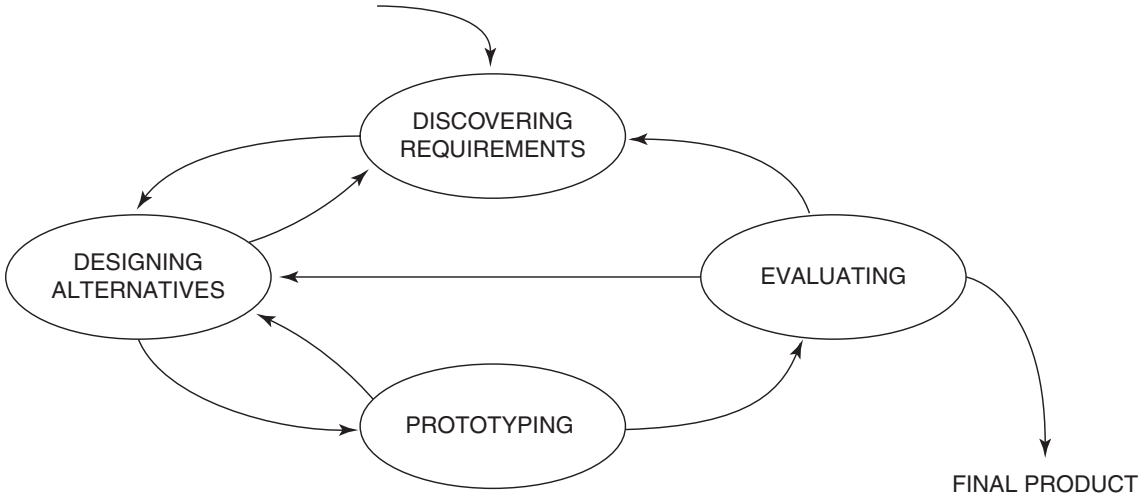
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# Lesson topics

- I will present you some **tips for data collection** and ask you to put them in practice by doing some exercises
  - Some of these exercises are **individual**, some are meant to be conducted **in group**
  - Exercises are mainly using **pen and paper.** At the very end, I will ask you to take your **laptop/tablet**
- We will have some hands-on activity on two main data collection methods:
  - **Qualitative**: semi-structured interviews
  - **Quantitative**: questionnaires

# Recap

- We are in the **discovery requirement phase** of the Interaction Design process




- We want to design a solution that improves users' lives
- To conceive an appropriate, usable, and desirable solution **we need first to understand what is the situation <u>right now</u>**
- Therefore, we need to conduct **research with users**

# Study design

- We need both **interviews** and **questionnaires** to have a study as much informative as possible
  - The combination of techniques is an important factor
  - Note that other techniques (e.g., observations) are also important
- We need to find the **right questions** to ask
  - For this purpose, we need set one or a few **goals** for the data collection: **what do we want to know?**
- How can we know what to ask?
  - We need to step back and apply the **PACT analysis** framework to **frame the domain properly**

# Exercise 1: PACT analysis

- Outline the people, activities, contexts, technologies (in the wide sense of all the tools used) that **at the moment** characterize the design problem for your project

- If you do not have a project topic yet, conduct a PACT analysis on this design brief:
  - *"you are asked to design an intelligent wearable device that supports beginners to learn the sport of skiing"*

# Exercise 1: conduct the PACT analysis

- Phase 1
  - **Individual** exercise
  - Pen and paper
  - 10 minutes
  - If the domain you are designing for is not familiar to you and you cannot define a PACT analysis for it, do desk research (i.e., search for information, e.g. read online documents or do benchmarking analysis)
- Phase 2
  - **Compare** your PACT analysis with your group members
  - If they differ, find an agreement
  - 5 minutes

# Interviews: pros and cons

**Pros**

- Very informative, provide many of insights
- Help explore a topic that we don't know very well
- Can be adjusted on the go

**Cons**

- Require a lot of time to be conducted and analyzed
- May be dispersive

# Semi-structured interviews

- Are explorative, they are good to be used **when you want to know more** about a problem, a category of people, a phenomenon
- They usually answer better to **WHY questions** regarding motivations and goals, e.g.
  - **Why** did you chose Computer Science rather than Chemistry?
  - **How do you feel** when you swim in the swimming pool? **Why?**

# Interview content and structure

- We need to find the **right questions** to ask in relation to the problem we want to solve
- We have done the PACT analysis, so ground the questions on it, based on your research goals
- Take a sheet of paper and draw a **table**:
  - On each cell of the **left column**, write the investigation theme and your goals, i.e., the theme you will address with that group of questions and what do you want to know
  - On the corresponding **right cell**, write all the questions related to that investigation theme

# Interview content and structure

- Take a sheet of paper and draw a table:

| Themes & goals | Questions |
|---|---|
| Personal data | • Sex: M/F/prefer not say……<br>• Age: ….<br>• Title of study…. |
| People motivations and goals for their activities | |
| Habits | |
| Difficulties | |

# Interview content and structure

- Consider to structure the interview according to different phases:
  - **Intro**: you explain who you are and the purpose of the interview
    - Remember to provide the consent form
  - **Warm up**: easy questions (e.g., personal data, "how long have you been…", "what do like of…")
  - **Main body of questions**: …
  - **A cool-off moment**: "Ok, we are almost done, is there anything that came to your mind and you would like to add?"
  - **Closure**: "Thank you for your time! If you would like to see the results of this work we might keep in contact and ask you to try our first prototype".

# Good and bad questions

- **Avoid biased questions: be careful with wording**, e.g.

  - *Has your experience of taking the bus ~~**improved**~~ after the electronic display at the bus stop was installed?*

**vs.**

  - *Did your experience of taking the bus has **changed** after the electronic display at the bus stop was installed?*
  - *If yes, **how**?*

# Good questions

- You need to understand the current practices and strategies people put in action to overcome problems. A good way to do it is to ask for **anecdotes**:

  - **Routines**: *"Could you please tell me what is your typical trip for getting to work in the morning?"*

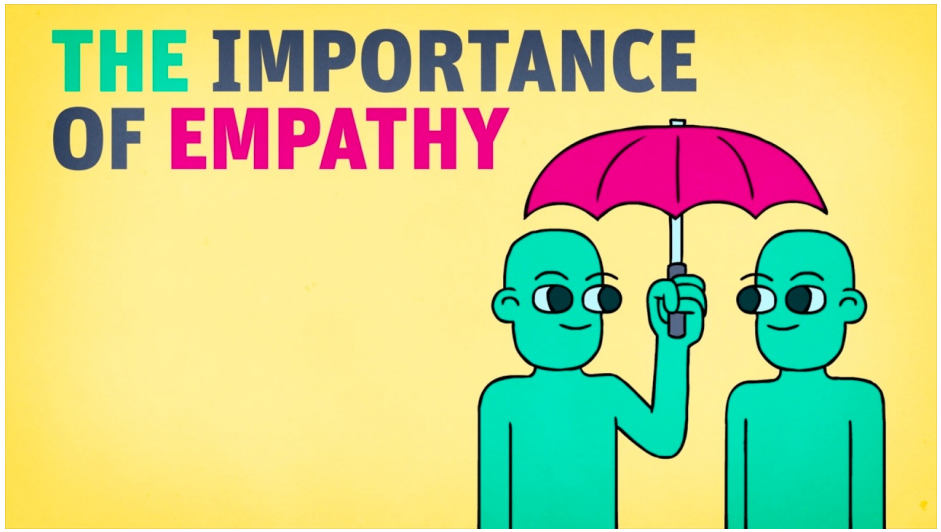
  - **Breakdown episodes**:
    - *"Can you recall of last time that you lost the bus and you were in a hurry? What did you do?"*
    - *"Can you recall of a time you had a problem of understanding with a teammate during an important sport competition? What happened?"*

# Organizational aspects

- **Calibrate the duration of your interview** according to where you want to conduct it and how much time do you want it to take (e.g., on the bus vs. sitting in a café)
- For long interviews, consider recording (audio, video) and then transcribing the most important parts or consider being in **2 people**, an interviewer and a note taker (in this case be very kind and empathic in order not to make the interviewee feel under exam)
- Note for the exam: you must demonstrate to have conducted the interviews (e.g., provide in the Google Drive folder the transcriptions, videos, audios, recording of skype calls, etc.)

# Attitude

- **You are not the expert**, even if you could be a potential target user as well
- You need to act as an alien and **ask the questions as you don't know <u>anything</u>** (be impassive)
- At the same time being **empathic** with the interviewee, i.e., kind, listen, understand



# Must-do: conduct a pilot

- Take someone, whoever (siblings, house mate..) and make a simulation of the interview
- Only by conducting a pilot, you will realize:
  - If the questions are clear
  - If you get the information you want
  - How long it takes to conduct the interview
  - You take confidence in asking the questions
- Don't use those results for your main analysis, you need to perform new interviews with different people, after you have amended the original interview structure into the final version
- Note for the exam: remember to indicate in the report that you conducted a pilot test and what you learned from it

# Exercise 2: interviews design

- In groups
- 20 minutes
- Write the **protocol interview** of your design project by following the tips I explained before
- Those who have not yet a project topic, can work on the case study of skiing:
  - *"you are asked to design an intelligent wearable device that supports beginners to learn the sport of skiing"*

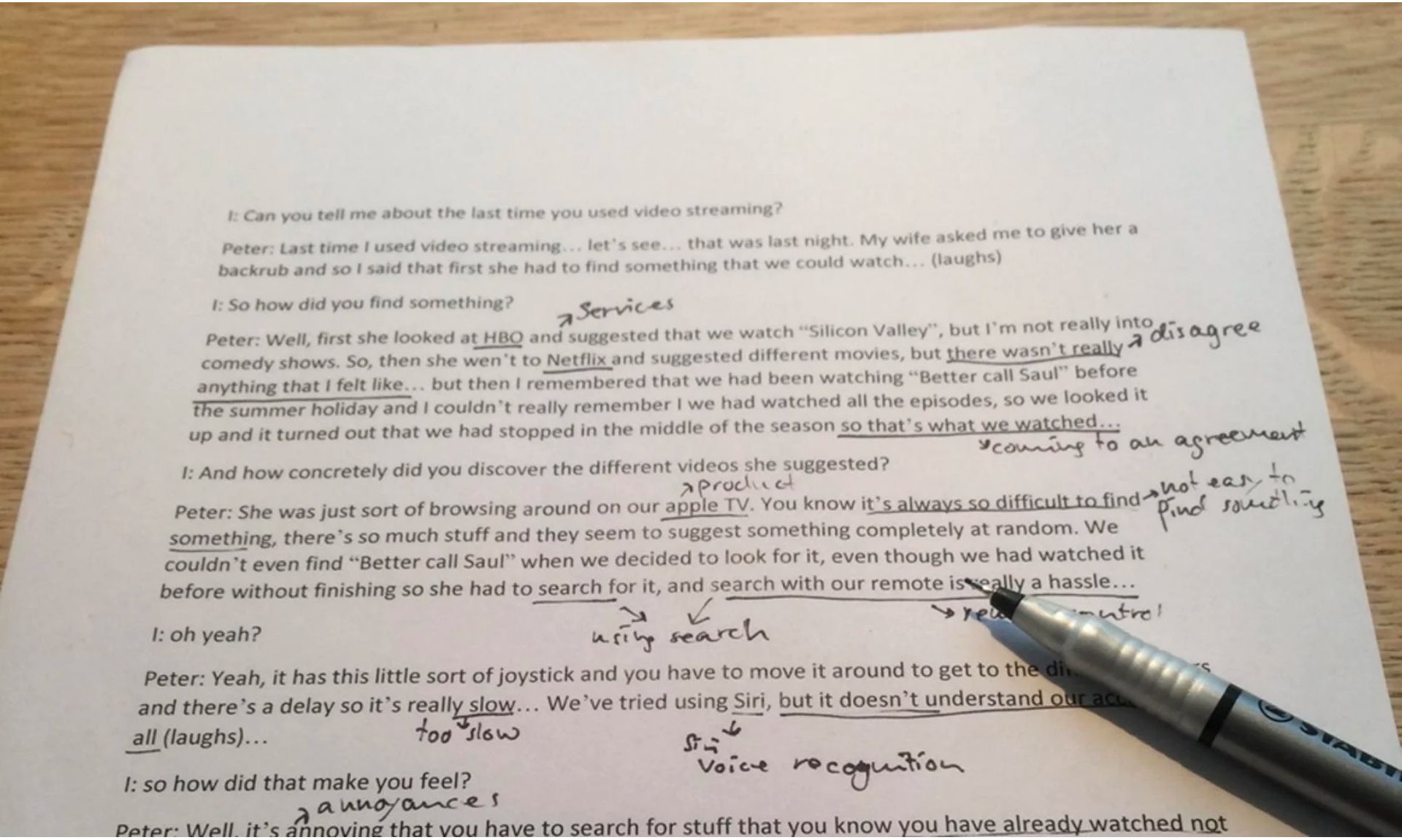
# Interview analysis: Thematic Analysis

- Thematic analysis is the most common form of analysis in qualitative research
- **Themes are patterns** across datasets that are important to the description of a phenomenon (and in general are associated to a specific **research question**)
- The thematic analysis process follows a systematic approach, resulting in a collection of themes that provide an accurate representation of **user needs, motivations, and behaviors**

# How to conduct an inductive thematic analysis

- Read the transcription of the interviews or the notes and **search for recurring patterns (themes)**
- Put a **code** (i.e., label) next to each interesting thing you notice
- From the initial codes, try to identify themes and families of themes. E.g.,
  - Family theme: Motivation
  - Themes: all the different motivations that interviewees have reported
- **Compare** your analysis with that of your partners and find an **agreement** on the codes
- Remember that you are looking both for **insights** and **recurrences**, e.g., a problem that is for many people, **but also one person describing an important situation is valuable**

# Generating initial codes



# Looking for themes

- Next phase: **look for potential themes from the codes**
- A **theme** is a recurring pattern of codes that captures something significant **about the original research question**
- E.g., we have a series of codes highlighted such as *HBO*, *Netflix*, and *Hulu;* the research question is based around video streaming so we might pick a theme called *"important streaming services"*

# Example: from insights, to codes, to themes

| User #1 | Codes | Themes |
|---|---|---|
| Juggle so many balls | time management, job duties | |
| Trying to get to an all in one platform | consolidation, ease of use | |
| Need to be compliant | analytics story | |
| Needs to run an operation | analytics story | |
| Ability to take what we are doing and look at a storyboard | analytics story | |
| Reduce cognitive load | time management, reduce complication | |
| great reports but we want to be able to "drag and drop" modules so we can manipulate them | wants to manipulate data | |
| We have 36 customers so a standard report may not be possible | customization | |
| Wants to have more control (my observation) | more control over the data | |
| All the data is there - but have to manipulate it herself | data available, has to customize | |
| Something that allows her to build her own story | wants to build a story | |
| Needs outcome data | outcomes | |
| "how many times did they administer TPA" | data story | |
| "how many times did they admit a psych patient" | data story | |
| "for anyone placed on a 5150 hold" who wasn't notified? | data story | |
| wants the journey of a visit and outcomes | data story | |
| "you are holding patients because it takes six hours to activate us" This can be taken to a c-suite | data story | |
| choose to customize an output when drilling in | customization | |
| looks at the information monthly | information availability | |
| some people like tammy look daily | information availability | |
| "how many cases" "were they open/closed" | data story | |
| trying to justify existence to c-suites | justification, justify expense, justify program | |
| c-suites want to know outcomes. they want to know the impact. | outcomes, impacts | |
| She would be able to design her own and have her own portal and make her own dashboards | customization of data and dashboards | |

# Find patterns among interviewees

- Understand how many participant report sentences that might be related to that theme: **count them**

| A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|
| | P1 | P2 | P3 | P4 | P5 | P6 |
| User is annoyed by the slow login process | | | | | | |
| Uses the "Calculate" button | | | | | | |
| Says something positive about the calculator feature unprompted | | | | | | |
| Asks for help with finding the "Management" page | | | | | | |
| Expected the process of adding a new contact to be much quicker | | | | | | |
| Discovered the voice feature | | | | | | |
| Understands the meaning of submitting a lead request | | | | | | |
| Surprised (in a bad way) by the results page of the lead search | | | | | | |
| The user experiences flow | | | | | | |
| Double-clicks icons (rather than single-clicks) | | | | | | |
| Confused by the last paragraph in the Lead page | | | | | | |
| Says that the Reports page has a clean look | | | | | | |
| Keep clicking the "visit now" text in the home page | | | | | | |
| Attempted submitting a lead request without filling in 2 mandatory fields | | | | | | |
| Thinks the product is relatively slow | | | | | | |
| Didn't notice the message bar | | | | | | |
| Complains about font size | | | | | | |

# Exercise 3: Thematic Analysis Case Study

- Individually
  - 10 minutes
  - Identify possible different themes in the short excerpt of interview I provide you in the next slide
  - Use a different code for each important part you identify
  - Create themes from the generated codes
- In group
  - 5 minutes
  - Check the consistency of coding: check your codes with the other group members and find an agreement

# Exercise 3: Case study "working in a library"

- **P1:** "I love working in Library A because it is such a lovely place to work, near the park. The difficulty here is I hate working in Library B, I think it's a slum; it's an airport lounge. I can't stand Library C which is even more of a slum; I don't care for Library D very much, so I don't like working there. Working in Library E there is very pleasant and I enjoy that, although you don't get much done because people come and talk to you and show you things, there's chat and you can't do long stints. At Library E, at 3.45 they ring a bell and you have to go and have tea".

- **P2:** "I love Library A to work in, as there was always silence and a good air conditioning. On the contrary, it's a pain to get books out in Libraries B and C because there is a mess in the organization and seats and desks are old. It is a pleasure to work in library E, I think its just amazing as a building and I never really thought about the extent to which the environment (walls, lights, etc.) affects me positively.

# Exercise 3: [partial solution: insights]

- **P1:** "I love working in Library A because it is such a lovely place to work, near the park. The difficulty here is I hate working in Library B, I think it's a slum; it's an airport lounge. I can't stand Library C which is even more of a slum. Working in Library D there is very pleasant and I enjoy that, although you don't get much done because people come and talk to you and show you things.

- **P2:** "I love Library A to work in, as there was always silence and a good air conditioning. On the contrary, it's a pain to get books out in Libraries B and C because there is a mess in the organization and seats and desks are old. It is a pleasure to work in Library D although it is crowded and a bit noisy, I think its just amazing as a building and I never really thought about the extent to which the environment (walls, lights, etc.) affects me positively.

# Questionnaires : pros and cons

**Pros**

- Can rich a high number of respondents
- Quick to answer
- Answers across respondents are easy to compare
- Analysis of answers easier

**Cons**

- Can put ideas in respondent's head
- Respondents with no opinion might answer anyway
- Respondents can feel constrained and frustrated
- Cannot tell if respondent misinterpreted the question

# Questionnaire design considerations

- Aspects to care of when creating a questionnaire:
  - Questionnaire purpose (i.e., research **goal**)
  - **Participants selection**
  - Question content (**what** to ask)
  - Question wording (**how** to ask it)
  - Response format
  - Question sequence (sometimes is importance of **shuffling** questions to avoid order effects)
  - Questionnaire **length**
  - Questionnaire **layout**
  - **Piloting**

# Potential pitfalls

- Bad selection of participants
- Failing memory of participants
- Effect of fatigue, feeling bored, time pressure
- Social desirable replies, e.g.,
  - participant does not want to appear stupid
  - “please the researcher" effect

# Some tips for questionnaires

- Write a paragraph to present the questionnaire:
  - A questionnaire with a title and good layout is generally perceived to be more credible than one without
  - Clarify the purpose of the questionnaire. What the answers will be used for? **But do not add information that can bias the respondents!**
  - The affiliation of the researcher can make a world of difference (say that you are students from UniTn)
- Leave ample white space for comments. It also makes the questionnaire appear easier
- **Resist the temptation to add more and more questions…**

# Some tips for questionnaires

- Note: remember always to add the **consent form**
- You can ask participants to leave their contact (e.g., email, phone, skype ID, etc.), but you need to clarify that this information will not be made public in any way, and that this is an optional input
- You might use the contact e.g., to ask further questions or make interviews

# Before writing the questions

- Write down the **goals** of your questionnaire (what do you want to know from it)
- Where to share / advertise / administer it? Online or on paper?
- If online, where?
  - On a group chat on Whatsapp? (Will then people try to fill the questionnaire from their smartphones)?
- Option: a selective question to understand if the person is a member of our population study or not
  - E.g., beginners, intermediates or experts videogame players, may have a different set of questions

# Writing questions

- Use simple wording
- Be brief
- Be specific
- Avoid:
  - Using biased wording
  - Using abbreviations or scientific jargon

# Writing questions

- Questions should ask for **only 1 piece of information**, so avoid:
  - Asking two questions at once, e.g. *"Do you buy frozen meat and frozen fish?"* A *"Yes"* answer can mean the respondent buys meat or fish or both.
  - Asking questions that contain assumptions, e.g. "*Did you like product X?"* (assumption that respondent has bought at least once this product)
- Respondents are able to answer (seriously) only about past and present experiences, feelings, ideas
- **Avoid asking hypothetical questions**, e.g.
  - *If we are going to build in feature x, are you going to use it?*

# Questions wording

- Question wording should ensure that every respondent will be answering the same thing, so **avoid:**
  - **Ambiguous wording** or wording that might mean different things to different respondents
  - Using terms for which the definition can **vary** (if it is unavoidable, provide the respondent with a definition)
  - Being ambiguous about the **time period or place or other aspects** the respondent should consider

# Common issues

- Not resisting to ask many, **too many questions**
- Using lots and lots of **words that are familiar just to the researcher** (and maybe little to the respondents)
- Not performing a **pilot test** on a questionnaire

# Exercise 4: find the bias in the questions

- *How dumb is (insert politician) when it comes to foreign policy?*
- *Should concerned dog owners vaccinate their pets?*
- *Where do you like to party?*
- *How happy or unhappy are you with the rate of current school board funding and the common-core curriculum?*
- *Do you always shower before bed?*
- *How amazing was your experience with our customer service team?*
- *What problems did you have with the launch of this new product?*
- *Would you be worried if we discontinued this product line?*

# Exercise 4: [solution]

- *How dumb is (insert politician) when it comes to foreign policy?*
  - This immediately brings a negative connotation to the question. Instead, you might ask the question: *Please describe your politician's position on foreign policy.*
- *Should concerned dog owners vaccinate their pets?*
  - By using the word concerned, you put pet owners who don't vaccinate their pets on the defensive, thus creating bias. Instead, ask it this way: *Do you think dogs should be required to be vaccinated?*
- *Where do you like to party?*
  - Well, what if the respondents don't like to party? What if they are homebodies? Instead, you could ask it like this: *What do you like to do on weekend evenings?*

# Exercise 4: [solution]

- *How happy or unhappy are you with the rate of current school board funding and the common-core curriculum?*
  - This is asking a lot of your respondents. Some might answer both questions, but many others will concentrate on the one that means the most to them. Instead, you could ask it like this: *How happy or unhappy are you with the rate of current school board funding?* And, next question: *What do you think of the common core?*

- *Do you always shower before bed?*
  - The obvious answer for most people is no. You've basically backed respondents into a corner. But, what if you sell shower gel, and you want to know specific showering habits for marketing purposes? This is where you want to offer respondents some options. You might ask the question this way: *How many nights a week do you shower before bed?*

# Exercise 4: [solution]

- *"How amazing was your experience with our customer service team?"*
  - This question is set up in a way that you've assumed already you thought the customer service team was amazing, you've left no room for another answer. The customer is now obliged to rate the customer service team on a scale of how amazing they were.

- *"What problems did you have with the launch of this new product?"*
  - This question assumes that there was something wrong in the first place and will have the customer looking for problems in their answer.

- *"Would you be worried if we discontinued this product line?"*
  - This example is using emotional language to lead a user: it suggests to the respondent that they *should* be worried, simply because the emotional language is in there.

# Google Forms

- https://docs.google.com/forms
- On Google forms it is possible to:
  - Create sections
  - Link sections to specific answers (e.g. if the user answers "cats" as preferred animal, go to section 4 where we will ask the questions about cats)
  - Shuffle questions
  - Add images (e.g., you can use the Self-Assessment Manikin)
- On a multiple choice question, please consider adding "OTHER" (when relevant) and the possibility to fill it in by the users so to explain it

# Exercise 5: create a questionnaire on Google Forms

- Explore the features and play around with the options
- Create at least 5 different questions
- Send the link to your group members and ask them to fill it in
- Export the results

# Bibliography


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